

PSG COLLEGE OF TECHNOLOGY, COIMBATORE - 641 004

Department of Computer Applications

MCA - I

OPEN BOOK TUTORIAL 4

23MX14 – DATABASE MANAGEMENT SYSTEM

Date: 11/12/2024

1.A) State the Properties of Transactions. What problems can happen if concurrency is not controlled during transaction processing?

[3]

B) i) Consider a relation R with six attributes ABCDEF with the following FDs, (A,C) $\rightarrow$ BDEF, B $\rightarrow$ EF, and C $\rightarrow$ BD. Is relation R in 3NF? Justify your answer. [5]

ii) How can you use grant, revoke and view concept to provide a secure access to other users? Give an example.

[5]

C) Explain the purpose of normalization process. Consider a company stores employee records in the format shown below. Here they show two such employees.

Employee ID: E05 Name: Fred Shred Job Title: Manager		
Branch Code: B04 Branch Name: London Branch Ph: 45432 Salary: 25,000		
Qualification	Level	Year Obtained
BSc	Undergraduate	1986
MSc	Postgraduate	1990

Employee ID: E07 Name: Jim Brown Job Title: Assistant		
Branch Code: B04 Branch Name: London Branch Ph: 23432 Salary: 15,000		
Qualification	Level	Year Obtained
BSc	Undergraduate	1995

i) Identify the insertion, deletion and modification anomalies associated with this relation.

ii) Identify the functional dependencies, state your reasonable assumptions if necessary.

iii) Identify the repeating group of attributes and transform the above unnormalised table into tables that are in 1st Normal Form.

iv) Identify any partial dependencies and transform into tables that are in 2nd Normal Form.

v) Identify any transitive dependencies and transform into tables that are in 3rd Normal Form.

2.i) Explain the various methods to implement Join operations using SQL statements. Consider the following two relations Student and Course. Write relational algebra statements and equivalent SQL to perform inner join and all the outer join operations on Student and Course, also write the resultant relation of each.

[12]

Student Id	Student name	Course Code
S1	A	L7
S2	B	L1
S3	C	Null

Course code	Course name	credit
L7	DBMS	4
L2	OS	3
L1	DM	5

(OR)

ii) Explain the purpose of following clause of SQL select statement with an example. “SELECT___FROM___WHERE___ Group by___ Having___ Order by”. Explain the purpose of Indexing and Triggers in database with examples. [12]

3. Consider a social network database, about people and their relationships. The database has two relations:

Emp_id in Children relation act as foreign key to refer Emp_id in Employee relation. Write the SQL statements that define the relational schema for this database.

EMPLOYEE

Emp_id	Name	Department	Address
D1	M. Raj	Development	Coimbatore
D2	B. Mani	Development	Erode
D3	D. Selvam	Sales and Purchase	Salem

CHILDREN

First_name	Emp_id	Age	Sex
Arun	D1	14	M
Krishna	D1	7	F
Senthil	D3	10	M
Achu	D2	5	F

Whether the user is able to perform the following operations in these relations?

Justify your answer.

i) Change the Emp_id's 'D1' into 'D2' in the Employee relation;

ii) Delete 'D. Selvam' tuple from the Employee relation;

iii) Insert new dependent with the following data.

(‘Raja’, ‘D4’, 17, ‘M’)

iv) Write a SQL query that computes, for each person, the total number of their relatives. Your query should return results containing the Emp_id, the name, and the count. [6]

4. Consider the relation *PLAYER* with relational schema *PLAYER* (*Player-no*, *Player-name*, *Team*, *Team-color*, *Coach-no*, *Coach-name*, *Player-position*, *Team-captain*) and set of functional dependencies as follows;

$F = \{Player-no \rightarrow Player-name, Player-no \rightarrow Player-position, Player-no \rightarrow Team, Coach-no \rightarrow Coach-name, Team \rightarrow Team\ color, Team \rightarrow Coach-no, Team \rightarrow Team-captain\}$

Answer the questions given below; a) Is

PLAYER in 2NF? If not, convert into 2NF. b) Is

PLAYER in 3NF? If not, convert into 3NF.